

On the intractability of approximating the antiferromagnetic Ising model

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Abstract: The problem of approximating the partition function of zero-field, antiferromagnetic Ising model in neighborhoods of its complex roots is considered. We demonstrate that the computational intractability of the problem is a consequence of the limitations of inductive inference and the conservation of complexity. Any solution to the problem would require the representation of an extensive amount of information and ensures its intractability.

MSC2020 subject classifications: Primary 03D15, 94A17, 60A10; secondary 82B26.

Keywords and phrases: Inference, Complexity, Intractability, Entropy, Ising model, Partition function.

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1. Introduction

The mathematical problem of approximating the partition function of the zero-field, antiferromagnetic Ising model on general graphs is regarded as computationally intractable under the current hypotheses and known algorithms of computational complexity theory. [8, 15, 3] We inquire on the preeminent measure of *inferential complexity* in our analysis of the approximation problem, instead, since the endeavor to solve any mathematical problem is one of reason that may or may not involve steps mechanized by an algorithm. We demonstrate that the *inferential intractability* of the approximation problem logically precedes and, therefore, implies its computational intractability.

As a function of the connected, simple graph $G = (V, E)$ with $n = |V|$ vertices and $m = |E|$ edges, the partition function Z of the zero-field Ising model is defined as

$$Z(G | \beta) = \sum_{\sigma \in \{-1, +1\}^n} \exp \left\{ \beta \sum_{uv \in E} \sigma_u \sigma_v \right\},$$

for some fixed interaction strength $\beta \in \mathbb{C}$, and proportional to the modified partition function

$$\mathcal{Z}(G \mid \zeta) = \sum_{y=0}^m \zeta^y \mu_0(y \mid G) \quad (1)$$

of the van der Waerden expansion with $\zeta = \tanh \beta$, [16, 9] where $\mu_0(y \mid G)$ is the probability that a random spanning subgraph of G consists of y edges and is a cycle. The joint probability distribution $\mu(x, y \mid G)$ of the number X of vertices of odd degree and number Y of edges in a random spanning subgraph of G specifies the probability $\mu_0(y \mid G) = \mu(0, y \mid G)$ and, thus, encodes the minimally sufficient information from G for solving the problem of approximating the modified partition function $\mathcal{Z}$.

We quantify the inferential complexity of a mathematical problem that involves a system by the information rank of the problem: the number of minimally sufficient statistics of the system required to solve the problem. The mathematical problem is inferentially intractable to an observer, if the rank of the problem exceeds the limited information bandwidth of the observer. If the inferential complexity of a problem is extensive, that is, if the rank of the problem scales with the degrees of freedom of the system, then it will surely be intractable to any observer with a limited capacity for information representation.

The Ising model originated in statistical physics as is a two-state spin system with pairwise interactions across the edge set of a simple graph. [13, 1] The log-partition function $\log Z$ is a moment generating function of the Gibbs distribution associated with the statistical mechanical model and characterizes the statistical and thermodynamic properties of the Ising system as such. The analyticity of $\log Z$, in particular, is determinative of order-disorder phase transitions of the model, as well as its computational [12] and inferential intractability.

If the topology of a graphical ensemble forbids spatial mixing and promotes long-range correlations and non-uniqueness, then the locus of complex Fisher zeros of the partition function $Z(G \mid \beta)$, as a function of $\beta \in \mathbb{C}$, intersects the real axis and delivers a root $\beta_* \in \mathbb{R}$ of Z [11, 14] that corresponds to a real root $\zeta_* = \tanh \beta_*$ of $\mathcal{Z}$. The problem of approximating $\mathcal{Z}$ is NP-hard, in particular, on d -regular graphs beyond the uniqueness threshold of the Gibbs measure on the d -regular infinite tree. [15] We have, therefore, elected to demonstrate the intractability of approximating $\mathcal{Z}$ on Ramanujan graphs in a neighborhood of the real root $\zeta = \zeta_0$ in order to resolve the P versus NP problem by demonstrating $\text{NP} \not\subseteq \text{P}$.

The computational intractability of the problem of approximating the modified partition function $\mathcal{Z}$ on Ramanujan graphs follows from its inferential intractability and the limitations of inductive inference. We demonstrate that the problem of approximating the modified partition function $\mathcal{Z}$ on a Ramanujan graph in neighborhoods of complex zeros of $\mathcal{Z}$ requires an exponential number of sufficient statistics to represent the relevant random process on the graph G . An epistemic horizon appears as a signal-to-noise ratio collapse in this context for any observer of the problem with a limited information bandwidth.

We begin with the moment decomposition of a discrete probability distribution that provides the infrastructure for representing the information the probability distribution encodes.

2. The moment decomposition

Let P be a discrete probability distribution with support $S \subseteq \mathbb{Z}$ of size $n = |S|$ and contained by $S' \subseteq \mathbb{Z} = \{0, 1, \dots, n' - 1\}$, for some $n' \geq n$, and suppose that the functions $f_j : \mathbb{Z} \mapsto \mathbb{R}$ satisfy

$$P(j) = \sum_{k \in S'} f_j(k) P(k), \quad (2)$$

for $j \in S'$. Let $\mathbf{p} \in \mathbb{R}^{n'}$ have components $p_j = P(j)$ and define $\mathbf{F} \in \mathbb{R}^{n' \times n'}$ to have components $F_{jk} = f_j(k)$, for $j, k \in S'$, so that $\mathbf{p} = \mathbf{F}\mathbf{p}$.

2.1. The Newton transform

Let $\mathbf{B} \in \mathbb{Z}^{n' \times n'}$ be the Newton matrix [4] with components $B_{kh} = \binom{h}{k}$, for $k, h \in S'$, and define $\mathbf{C} = \mathbf{F}\mathbf{B}^{-1}$. The probability vector $\mathbf{p} = \mathbf{C}\mathbf{e}$ is, thus, a linear transformation of the moment vector $\mathbf{e} = \mathbf{B}\mathbf{p}$ with components $e_k = E_P\left(\binom{Z}{k}\right)$, for $k \in S'$. The matrix $\mathbf{C}$ has components

$$\begin{aligned} c_{jh} &= \sum_{k \in S'} (-1)^{h-k} \binom{h}{k} f_j(k) \\ &= \Delta^h f_j(0) \end{aligned}$$

is the h th forward difference operator applied to f_j at zero, for $h, j \in S'$.

Proposition 1. *There exists a unique function $f_j : \mathbb{Z} \mapsto \mathbb{R}$ that satisfies Equation (2) and the linear constraints $|c_{jh}| \geq |c_{j(h+1)}|$, for $j, h \in S'$ such that $h < n' - 1$.*

Definition 1. The function f_j satisfying Proposition 1 is the *order function* of the individual probability $P(j)$, for $j \in S'$.

Remark 1. The order function f_j is the lowest-order polynomial interpolating the points $(k, f_j(k))$, for $k \in S$ and $j \in S'$.

Definition 2. The matrix $\mathbf{F} \in \mathbb{R}^{n' \times n'}$ is the *order matrix* of the probability distribution P , with rows of order functions $f_j : S' \mapsto \mathbb{R}$, for $j \in S'$.

Proposition 2. *The individual probability $P(j)$ is uniquely expressed as the linear combination*

$$P(j) = \sum_{k \in S'} c_{jk} E_P\left(\binom{Z}{k}\right)$$

of the moments $E_P\left(\binom{Z}{k}\right)$, for $j, k \in S'$.

2.2. Low-rank approximation of P

Let $\hat{P}$ be an approximation to P and define $\hat{\mathbf{p}} \in \mathbb{R}_+^{n'}$ and $\hat{\mathbf{F}} \in \mathbb{R}^{n' \times n'}$ analogously to $\mathbf{p}$ and $\mathbf{F}$, respectively, and define $\hat{\mathbf{C}} = \hat{\mathbf{F}}\mathbf{B}^{-1}$ and $\hat{\mathbf{e}} = \mathbf{B}\hat{\mathbf{p}}$ in a corresponding manner. The difference $\hat{\mathbf{p}} - \mathbf{p}$ of the approximate and exact probability vectors decomposes as

$$\begin{aligned}\hat{\mathbf{p}} - \mathbf{p} &= \hat{\mathbf{C}}\hat{\mathbf{e}} - \mathbf{C}\mathbf{e} \\ &= \hat{\mathbf{C}}(\hat{\mathbf{e}} - \mathbf{e}) + (\hat{\mathbf{C}} - \mathbf{C})\mathbf{e}.\end{aligned}$$

Proposition 3. *If $\hat{e}_k = e_k$ for $k = 0, 1, \dots, r$ and $\hat{e}_{j_k} = 0$ for $j \in S'$ and $k = r+1, r+2, \dots, n'-1$, for some $0 \leq r \leq n'-1$, then $\hat{\mathbf{p}} - \mathbf{p} = (\hat{\mathbf{C}} - \mathbf{C})\mathbf{e}$.*

Definition 3. A probability distribution $\hat{P}$ satisfying the conditions of Proposition 3 is a *rank r approximation* of the probability distribution P .

Remark 2. Since $\hat{\mathbf{C}}\mathbf{e} = \hat{\mathbf{F}}\mathbf{p}$, then the order matrix $\hat{\mathbf{F}}$ of the rank r approximating distribution $\hat{\mathbf{p}}$, with order functions f_j for $j \in S'$, is the matrix of the linear transformation from the true distribution $\mathbf{p}$ to the rank r approximation $\hat{\mathbf{p}} = \hat{\mathbf{F}}\mathbf{p}$.

The rank r approximation $\hat{P}$ conserves the moments $E_{\hat{P}}\left(\binom{Z}{k}\right) = E_P\left(\binom{Z}{k}\right)$, for $k = 0, 1, \dots, r$, but is restricted from encoding any information related to higher-order moments of P . The principle of maximum entropy asserts the existence and uniqueness of $\hat{P}$, [5, 6] and prescribes it as a member of the exponential family of probability distributions. [17] The orthogonality of such a low-rank projection corresponding to the maximization of a relative entropy objective function is evident in light of the preceding material.

2.3. The exponential family representation of $\hat{P}$

Let $Q_r(z \mid \boldsymbol{\theta})$ be the member of the exponential family of probability distributions of rank $0 \leq r \leq s-1$ with sufficient statistics $t_k(z) = \binom{z}{k}$, for $k \in S'$, defined as

$$Q_r(z \mid \boldsymbol{\theta}) = \begin{cases} Q_0(z) \exp \left\{ \sum_{k=1}^r \theta_k t_k(z) - \psi(\boldsymbol{\theta}) \right\} & \text{for } z \in S \\ 0 & \text{otherwise,} \end{cases}$$

where $\boldsymbol{\theta} = (\theta_1 \ \theta_2 \ \dots \ \theta_r)' \in \mathbb{R}^r$ and $\psi(\boldsymbol{\theta})$ is the cumulant function

$$\psi(\boldsymbol{\theta}) = \log \sum_{z \in S} Q_0(z) \exp \left\{ \sum_{k=1}^r \theta_k t_k(z) \right\}$$

of the probability distribution. The function $Q_0 : S \mapsto \mathbb{R}_+$ is the base measure of the distribution and the parameters $\theta_k \in \mathbb{R}$ are Lagrange multipliers in the

constrained optimization problem:

$$\begin{aligned} & \text{minimize} \quad \text{KL}(p||q) \text{ over } p \in \mathcal{P} \\ & \text{subject to} \quad E_p\left(\binom{z}{k}\right) = E_P\left(\binom{z}{k}\right), \text{ for } k = 0, 1, \dots, r, \end{aligned}$$

where $\mathcal{P} = \{p : S \mapsto \mathbb{R}_+\}$ is the set of functions with support S . The solution $Q_r(z | \theta)$ is unique [7] up to the choice of base measure Q_0 that encodes the prior knowledge of the observer on the probability distribution P .

Proposition 4. *Let $\hat{\mathbf{F}}_r \in \mathbb{R}^{n' \times n'}$ be the order matrix of $Q_r(z | \theta)$. The coefficient matrix $\hat{\mathbf{C}} = \hat{\mathbf{F}}_r \mathbf{B}^{-1}$ has components $c_{jk} = 0$, for $k > r$.*

The probability distribution $Q_r(z | \theta)$ satisfies the conditions of Proposition 3 and, therefore, the difference of the individual probabilities is expressed as

$$Q_r(j | \theta) - P(j) = \sum_{h \in S'} (\hat{c}_{jh} - c_{jh}) E_P\left(\binom{Z}{h}\right),$$

for $j \in S'$. The orthogonality of the residual weights $\hat{c}_{jh} - c_{jh}$ to rows of the Newton matrix is implied by the equality

$$\sum_{j \in S} \binom{j}{k} [Q_r(j | \theta) - P(j)] = \sum_{h \in S'} E_P\left(\binom{Z}{h}\right) \sum_{j \in S'} \binom{j}{k} (\hat{c}_{jh} - c_{jh}),$$

from which it follows that $0 = \sum_{j \in S} \binom{j}{k} (\hat{c}_{jh} - c_{jh})$, for $h \in S$ and $0 \leq k \leq r$.

3. The epistemic horizon

The modified partition function $\mathcal{Z}(G | \zeta)$, as defined in Equation (1), may be expressed otherwise as

$$\mathcal{Z}(G | \zeta) = \sum_{y=0}^m \zeta^y \mu(y | G) \sum_{x \in S_y} f_y(x) \mu(x | y, G),$$

for any function $f_y : \mathbb{Z} \mapsto \mathbb{R}$ such that

$$\mu(0 | y, G) = \sum_{x \in S_y} f_y(x) \mu(x | y, G),$$

where $S_y \subseteq \{0, 1, \dots, n\}$ is the support of the conditional distribution $\mu(x | y, G)$ and $\mu(y | G) = \binom{m}{y} / 2^m$ is the marginal distribution of Y . Let $f_y(x)$ be the order function associated with $\mu(x | y, G)$, in which case,

$$\mu(0 | y, G) = \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{h} \middle| Y = y \right],$$

where $s_y = |S_y|$ and the values $c_{yh} \in \mathbb{R}$ are those corresponding to f_y according to Proposition 3, for $h = 0, 1, \dots, s_y - 1$ and $y = 0, 1, \dots, m$. The modified partition function is, thus, represented in terms of moments of $\mu(x, y | G)$ as

$$\mathcal{Z}(G | \zeta) = \sum_{y=0}^m \zeta^y \mu(y | G) \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{h} \middle| Y = y \right].$$

3.1. Low-rank approximation of $\mathcal{Z}$

Let $\hat{\mu}_{\mathbf{r}}(x, y | G)$ be a rank $\mathbf{r} = (r_0 \ r_1 \ \dots \ r_m)' \in \mathbb{Z}_+^{m+1}$ approximation to $\mu(x, y | G)$, with respective rank $0 \leq r_y \leq s_y - 1$ approximations $\hat{\mu}_{r_y}(x | y, G)$ to the conditional distributions $\mu(x | y, G)$, for $y = 0, 1, \dots, m$. With the understanding that $\hat{c}_{yh} = 0$ for $h > r_y$, the approximation

$$\hat{\mathcal{Z}}_{\mathbf{r}}(G | \zeta) = \sum_{y=0}^m \zeta^y \mu(y | G) \sum_{h=0}^{s_y-1} \hat{c}_{yh} E_{\hat{\mu}_{r_y}} \left[\binom{X}{h} \middle| Y = y \right]$$

of $\mathcal{Z}(G | \zeta)$, as a consequence of Proposition 3, the relative error

$$\mathcal{E}_{\mathbf{r}}(G | \zeta) = \left| \frac{\sum_{y=0}^m \zeta^y \mu(y | G) \sum_{h=0}^{s_y-1} (\hat{c}_{yh} - c_{yh}) E_\mu \left[\binom{X}{h} \middle| Y = y \right]}{\sum_{y=0}^m \zeta^y \mu(y | G) \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{h} \middle| Y = y \right]} \right|, \quad (3)$$

for any simple graph $G = (V, E)$ and $\zeta \in \mathbb{C}$.

3.2. The exponential family representation of $\hat{\mathcal{Z}}$

In the absence of prior information on the topology of G , we project the set $\mathcal{D}_n \subseteq \mathbb{Z}_+^n$ of graphic vectors on n vertices [2] onto the set $S \subseteq \mathbb{Z}^2$ via the transformation $x = \sum_{w \in V} (d_w \bmod 2)$ and $y = \sum_{w \in V} d_w / 2$ for $\mathbf{d} \in \mathcal{D}_n$. Let $S'_y = \{(x', y') \in S \mid y' = y\}$ and define the rank r_y approximation to the conditional distribution $\mu(x | y, G)$ as

$$\hat{\mu}_{r_y}(x | \boldsymbol{\theta}_y, G) = \begin{cases} \hat{\mu}_0(x | y, G) \exp \{ \theta_0 t_0(x) + \sum_{k=1}^{r_y} \theta_{yk} t_k(x) - A(\boldsymbol{\theta}_y) \} & \text{for } x \in S'_y \\ 0 & \text{otherwise,} \end{cases}$$

where $\boldsymbol{\theta}_y = (\theta_{y1} \ \theta_{y2} \ \dots \ \theta_{yr_y})' \in \mathbb{R}^{r_y}$, for $y = 0, 1, \dots, m$, and $A(\boldsymbol{\theta})$ is the cumulant function of the distribution and $\hat{\mu}_0(x | y, G) \propto \binom{n}{x}$ is base measure since X is a sum of the binary random variables.¹

The parameters $\theta_{yk} \in \mathbb{R}$ are the Lagrange multipliers that ensure $E_{\hat{\mu}_{r_y}}[t_k(X) | Y = y] = E_\mu[t_k(X) | Y = y]$, for $k = 0, 1, \dots, r_y$ and $y = 0, 1, \dots, m$. The parameter $\theta_0 \in \mathbb{R}$ conserves the proportion of cycles $\sum_{y=0}^m \hat{\mu}_{r_y}(0 | \boldsymbol{\theta}_y, G) \mu(y | G) = 1/2^{n-1}$ in G [2] and binds the family of conditional distribution approximations.

¹See Remark 3.

3.3. Equivalence classes of moment motifs

Since $X = \sum_{w \in V} x_w$ is a sum of the random variables $x_w \in \{0, 1\}$, each indicating the parity of vertex $w \in V$ in a random spanning subgraph of G , then the binomial moment $E_\mu \left[\binom{X}{k} \middle| Y = y \right]$ decomposes as the sum

$$E_\mu \left[\binom{X}{k} \middle| Y = y \right] = \sum_{\substack{W \subseteq V \\ |W|=k}} E_\mu \left[\prod_{w \in W} x_w \middle| Y = y \right] \quad (4)$$

of product moments $E_\mu [\prod_{w \in W} x_w | Y = y]$, for $k = 0, 1, \dots, n-1$ and $y = 0, 1, \dots, m$.

Let $\mathcal{W}_k$ be the set of equivalence classes of vertex subsets $W \subseteq V$ of size $|W| = k$ under $M_y(W) = E_\mu [\prod_{w \in W} x_w | Y = y]$. The binomial moment in Equation (4) is succinctly represented as

$$E_\mu \left[\binom{X}{k} \middle| Y = y \right] = \sum_{W \in \mathcal{W}_k} N_W E_\mu \left[\prod_{w \in W} x_w \middle| Y = y \right],$$

where $N_W = |\mathcal{W}_k[W]|$ is the size of the equivalence class corresponding to $W \in \mathcal{W}_k$ and, in total, satisfy $\binom{n}{k} = \sum_{W \in \mathcal{W}_k} N_W$. The pseudo-randomness of the edge distribution of a Ramanujan graph, as guaranteed by the expander mixing lemma, requires conservation of moments corresponding to $\binom{n}{k}$ vertex subsets of size k to determine N_W , for $W \in \mathcal{W}_k$.

3.4. Combinatorial moments

The process of generating random spanning subgraphs of G conditional on the event $Y = y$ is equivalent to the process of collecting a random sample without replacement of y edges from the edge set of G and, therefore, the vector $\mathbf{x} \in \mathbb{Z}_+^n$ of vertex parities $x_w \in \{0, 1\}$ for $w \in W$ are statistically dependent. Nevertheless, this hypergeometric process is well approximated by a binomial process for large $y, m \in \mathbb{Z}_+$ and we consider this dependence negligible.

Let $M_y(W) = E_\mu [\prod_{w \in W} x_w | Y = y]$ and $\hat{M}_y(W) = E_{\hat{\mu}_{r_y}} [\prod_{w \in W} x_w | Y = y]$ for any $W \subseteq V$ and $y = 0, 1, \dots, m$. Although $\hat{M}_y(W) = M_y(W)$ for any $W \subseteq V$ such that $|W| \leq r_y$, no higher-order moment was conserved.² The differences in product moments $\hat{M}_y$ and M_y for $W \subseteq V$ with $|W| = r_y + 1$ is, therefore,

$$\hat{M}_y(W) - M_y(W) = \left[\hat{M}_y(w | W \setminus \{w\}) - M_y(w | W \setminus \{w\}) \right] M_y(W \setminus \{w\}),$$

where $M_y(w | W \setminus \{w\}) = E_\mu [x_w | \prod_{v \in W \setminus \{w\}} x_v = 1, Y = y]$ and $\hat{M}_y(w | W \setminus \{w\}) = E_{\hat{\mu}_{r_y}} [x_w | \prod_{v \in W \setminus \{w\}} x_v = 1, Y = y]$ for any $w \in W$ and $y = 0, 1, \dots, m$.

²See Section 4.1

By the law of total expectation, the conditional moment

$$E_{\hat{\mu}_{r_y}} \left[x_w \mid \prod_{v \in W \setminus \{w\}} x_v = 1, Y = y \right] = E_{\hat{\mu}_r} \left(E_{\hat{\mu}_r} \left[x_w \mid \prod_{v \in W \setminus \{w\}} x_v = 1, \mathbf{x}_{-W}, Y = y \right] \mid Y = y \right)$$

is an expected value, with respect to the rank r_y distribution $\hat{\mu}_{r_y}$ of the random vector $\mathbf{x}_{-W}$ with elements x_v for $v \notin W$, of a logistic regression with linear predictor

$$\eta_w(\mathbf{x}_{-w} \mid \boldsymbol{\theta}) = \sum_{k=1}^{r_y} \theta_k \mathbf{e}_{k-1}(\mathbf{x}_{-w}),$$

under the stipulation that $x_v = 1$ for $v \in W \setminus \{w\}$, where $\mathbf{e}_k : \{0, 1\}^s \mapsto \mathbb{Z}$ is the k th-order elementary symmetric polynomial for any $s \in \mathbb{Z}_+$.

The linear predictor, for example, with $r = 2$ and $W = \{w, u, v\}$, is a monotonic function

$$\eta_w(\mathbf{x}_{-w} \mid \boldsymbol{\theta}) = \theta_1 + \theta_2 \left(x_u + x_v + \sum_{s \in V \setminus W} x_s \right)$$

of x_u and x_v . The true statistical relationship among the triplet (x_w, x_u, x_v) under μ , however, is not monotonic in general: the moment $M_y(w \mid W \setminus \{w\})$ is over a probability distribution that encodes non-monotonic *exclusive or* relationships of the form $x_w = x_u + x_v - 2x_u x_v$ among the variables (x_w, x_u, x_v) . A similar argument generalizes to moments of higher order.

Proposition 5. *Let $W \subseteq V$ such that $|W| = r_y + 1$ be connected in G . There exists a constant $\gamma_{r_y}(G) > 0$ such that $|\hat{M}_y(W) - M_y(W)| \geq \gamma_{r_y}(G)$, for $r_y < s_y - 1$ and $y = 0, 1, \dots, m$.*

Proposition 6. *There exists a constant $\Gamma_{r_y}(G) > 0$ such that for a connected graph G ,*

$$\left| E_{\hat{\mu}_{r_y}} \left[\binom{X}{r_y + 1} \mid Y = y \right] - E_{\mu} \left[\binom{X}{r_y + 1} \mid Y = y \right] \right| \geq \Gamma_{r_y}(G),$$

for any $0 \leq r_y < s_y - 1$ and $y = 0, 1, \dots, m$.

3.5. Rank extensivity

In light of the expression for the relative error $\mathcal{E}_{\mathbf{r}}(G \mid \zeta)$ of the approximation $\hat{\mathcal{Z}}_{\mathbf{r}}(G \mid \zeta)$ in Equation (3), let $\epsilon > 0$ and define $\mathbf{r}_{\epsilon} \in \mathbb{Z}_+^{m+1}$ as the solution to the constrained optimization problem:

$$\begin{aligned} & \text{minimize} && R(\mathbf{r}) = 1 + \sum_{y=0}^m \sum_{k=1}^{r_y} \binom{n}{k} \\ & \text{subject to} && \mathcal{E}_{\mathbf{r}}(G \mid \zeta) \leq \epsilon \\ & && 0 \leq r_y \leq s_y - 1 \text{ for } y = 0, 1, \dots, m, \end{aligned}$$

over $\mathbf{r} \in \mathbb{Z}_+^{m+1}$, where the objective $R(\mathbf{r})$ is the number of statistics used in the rank $\mathbf{r}$ approximation $\hat{\mu}_{\mathbf{r}}$ of μ . The solution $\mathbf{r}_\epsilon$ corresponds to the least number $R_\epsilon = R(\mathbf{r}_\epsilon)$ of minimally sufficient statistics for approximating $\mathcal{Z}(G \mid \epsilon)$ with a relative error bounded by ϵ .

Let $\zeta_0 = \rho_0 e^{i\phi_0}$ be a root of $\mathcal{Z}(G \mid \zeta)$ and define the radius profile function

$$\begin{aligned} \mathcal{Z}_{\phi_0}(G \mid \rho) &= \sum_{y=0}^m e^{y \log(\rho/\rho_0)} \zeta_0^y \mu(y \mid G) \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{k} \middle| Y = y \right] \\ &= \log(\rho/\rho_0) \sum_{k=1}^{\infty} \log^{k-1}(\rho/\rho_0) \sum_{y=0}^m \frac{y^k}{k!} \zeta_0^y \mu(y \mid G) \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{k} \middle| Y = y \right] \end{aligned}$$

as a function of ρ with ϕ_0 fixed. The relative error of the approximation $\hat{\mathcal{Z}}_{\mathbf{r}}$ is, thus,

$$\mathcal{E}_{\mathbf{r}}(G \mid \rho e^{i\phi_0}) = \left| \frac{\sum_{k=0}^{\infty} \log^k(\rho/\rho_0) \sum_{y=0}^m \frac{y^k}{k!} \zeta_0^y \mu(y \mid G) \sum_{h=0}^{s_y-1} (\hat{c}_{yh} - c_{yh}) E_\mu \left[\binom{X}{k} \middle| Y = y \right]}{\log(\rho/\rho_0) \sum_{k=1}^{\infty} \log^{k-1}(\rho/\rho_0) \sum_{y=0}^m \frac{y^k}{k!} \zeta_0^y \mu(y \mid G) \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{k} \middle| Y = y \right]} \right|.$$

Unless $r_y = s_y - 1$ for $y = 0, 1, \dots, m$, the index $k = 0$ term in the numerator is nonzero and, crucially, is independent of the factor $\log(\rho/\rho_0)$.

In a complementary manner, define the angle profile function $\mathcal{Z}_{\rho_0}(G \mid \phi)$ as a function of ϕ with ρ_0 fixed as

$$\begin{aligned} \mathcal{Z}_{\rho_0}(G \mid \phi) &= \sum_{y=0}^m e^{i(\phi-\phi_0)y} \zeta_0^y \mu(y \mid G) \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{k} \middle| Y = y \right] \\ &= (\phi - \phi_0) \sum_{k=1}^{\infty} (\phi - \phi_0)^{k-1} \sum_{y=0}^m \frac{(iy)^k}{k!} \zeta_0^y \mu(y \mid G) \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{k} \middle| Y = y \right]. \end{aligned}$$

The relative error of the approximation, in this case, is expressed as

$$\mathcal{E}_{\mathbf{r}}(G \mid \rho_0 e^{i\phi}) = \left| \frac{\sum_{k=0}^{\infty} (\phi - \phi_0)^{k-1} \sum_{y=0}^m \frac{(iy)^k}{k!} e^{i\phi_0 y} \rho_0^y \mu(y \mid G) \sum_{h=0}^{s_y-1} (\hat{c}_{yh} - c_{yh}) E_\mu \left[\binom{X}{k} \middle| Y = y \right]}{(\phi - \phi_0) \sum_{k=1}^{\infty} (\phi - \phi_0)^{k-1} \sum_{y=0}^m \frac{(iy)^k}{k!} e^{i\phi_0 y} \rho_0^y \mu(y \mid G) \sum_{h=0}^{s_y-1} c_{yh} E_\mu \left[\binom{X}{k} \middle| Y = y \right]} \right|.$$

A similar conclusion may be drawn and, as a consequence, $\mathbf{r}_\epsilon \rightarrow \mathbf{s} - \mathbf{1}$ and $R(\mathbf{r}_\epsilon) \rightarrow (\mathbf{s} - \mathbf{1})$ as $\phi \rightarrow \phi_0$, where $\mathbf{s} \in \mathbb{Z}_+^{m+1}$ has components $s_y \in \mathbb{Z}_+$, for $y = 0, 1, \dots, m$.

Proposition 7. *The problem of approximating $\mathcal{Z}(G \mid \zeta)$ on Ramanujan graphs is inferentially intractable in a neighborhood of any of its complex roots.*

Proposition 8. *The problem remains inferentially intractable in the thermodynamic limit, i.e., $n \rightarrow \infty$, in a neighborhood around the implied real root $\zeta_0 < 0$ of $\mathcal{Z}$.*

4. Discussion

The complexity $S(G)$ of the problem of approximating the modified partition function $\mathcal{Z}$ is the sum $S(G) = H(G | \mathbf{r}) + K(G | \mathbf{r})$ of missing information $H(G | \mathbf{r})$ and known randomness $K(G | \mathbf{r})$ associated with the rank $\mathbf{r}$ approximation $\hat{\mu}_{\mathbf{r}}$ of μ . [18] With physical entropy characterized in this manner, as the sum of Shannon entropy and Kolmogorov complexity, Zurek salvaged the second law of thermodynamics from Maxwell's demon [10] by conserving the complexity of a system, whether or not observations of the system are recorded or represented.

The complexity of the system and, thus, uncertainty possessed by an observer of the approximation problem, is quantified as the Kullback-Leibler divergence $S(G) = \text{KL}(\nu || \hat{\nu}_0)$ from the probability distribution $\hat{\nu}_0(\mathbf{x}, y | G)$ that encodes the prior information known to the observer, to the joint probability distribution $\nu(\mathbf{x}, y | G)$ of the vector $\mathbf{x} \in \{0, 1\}^n$ of vertex parities and the number Y of edges in a random spanning subgraph of G .

Remark 3. The choice of prior distribution $\hat{\nu}_0(\mathbf{x} | G) \propto 1$ over $\mathbf{x} \in \{0, 1\}^n$ induces the prior probability distributions $\hat{\mu}_0(x | G) \propto \binom{n}{x}$ of $X = x_1 + x_2 + \dots + x_n$.

4.1. Sufficient statistics

The rank $\mathbf{r}$ approximation $\hat{\mu}_{\mathbf{r}}$ of μ is the projection

$$\hat{\mu}_{\mathbf{r}} = \arg \min_{\pi \in \Pi} \text{KL}(\hat{\nu}_{\mathbf{r}} || \pi),$$

where $\Pi = \{\pi : S \mapsto \mathbb{R}_+\}$ and $S = \times_{y=0}^m S_y$, of the probability distribution $\hat{\nu}_{\mathbf{r}}$ that solves the constrained optimization problem:

$$\begin{aligned} & \text{minimize} \quad \text{KL}(\omega || \nu_0) \text{ over } \omega \in \Omega \\ & \text{subject to} \quad E_{\omega} \left[\prod_{w \in W} x_w \mid Y = y \right] = E_{\nu} \left[\prod_{w \in W} x_w \mid Y = y \right] \\ & \quad |W| = k \text{ for } W \subseteq V \\ & \quad 0 \leq k \leq r_y \text{ for } y = 0, 1, \dots, m \\ & \quad P_{\omega}(X = 0) = 1/2^{n-1}, \end{aligned}$$

where $\Omega = \{\omega : \{0, 1\}^n \times \{0, 1, \dots, m\} \mapsto \mathbb{R}_+\}$. Each of the $R = 1 + \sum_{y=0}^m \sum_{k=0}^{r_y} \binom{n}{k}$ constraints in the optimization problem warrants a sufficient statistic in the exponential family representation of the probability distribution $\hat{\nu}_{\mathbf{r}}$ in order to conserve its respective moment.

4.2. Partitioning information

The quantity of information not encoded by the rank $\mathbf{r}$ approximation $\hat{\mu}_{\mathbf{r}}$ is the Kullback-Leibler divergence $H(G | \mathbf{r}) = \text{KL}(\nu || \hat{\nu}_{\mathbf{r}})$ and, thus, the information

encoded by the rank $\mathbf{r}$ approximation is the difference $K(G \mid \mathbf{r}) = S(G) - H(G \mid \mathbf{r})$. The missing information and known randomness are, respectively, expressed as

$$H(G \mid \mathbf{r}) = - \sum_{y=0}^m \sum_{\mathbf{x} \in \{0,1\}^n} \nu(\mathbf{x}, y \mid G) \log \frac{\hat{\nu}_{\mathbf{r}}(\mathbf{x}, y \mid G)}{\nu(\mathbf{x}, y \mid G)}$$

$$K(G \mid \mathbf{r}) = - \sum_{y=0}^m \sum_{\mathbf{x} \in \{0,1\}^n} \nu(\mathbf{x}, y \mid G) \frac{\hat{\nu}_{\mathbf{0}}(\mathbf{x}, y \mid G)}{\hat{\nu}_{\mathbf{r}}(\mathbf{x}, y \mid G)}.$$

Since $S(G) = H(G \mid \mathbf{r})$ then we arrive at the relationship $H(G \mid \mathbf{0}) = H(G \mid \mathbf{r}) + K(G \mid \mathbf{r})$, where $K(G \mid \mathbf{r}) = H(G \mid \mathbf{0}) - H(G \mid \mathbf{r})$.

4.3. Conservation of complexity

The complexity of the system $S(G) = H(G \mid \mathbf{0}) \in \Theta(n)$ is conserved throughout, whether or not the observer attempts to solve the approximation problem. An extensive quantity of information, however, is necessary to approximate the modified partition function $\mathcal{Z}$ in a neighborhood of a complex zero, thereby causing $H(G \mid \mathbf{r}_\epsilon) \in o(1)$ and $K(G \mid \mathbf{r}_\epsilon) \in \Theta(n)$.

The hypothesis that a tractable procedure exists for solving the approximation problem is refuted by a contradiction argument: Suppose that such a procedure exists. It would necessarily compress the minimally sufficient information $K(G \mid \mathbf{r}) \in \Theta(n)$ further so that $K(G \mid \mathbf{r}) \in o(n)$. The missing information is thus $H(G \mid \mathbf{r}) \in \Theta(n)$ and the problem of approximating $\mathcal{Z}$ in neighborhoods of its roots cannot be solved according to Proposition 7 and the conservation of complexity.

Remark 4. The contradictory argument is restored, however, if one were to dismiss the conservation of complexity and absurdly permit $S(G) > H(G \mid \mathbf{r}_\epsilon) + K(G \mid \mathbf{r}_\epsilon)$.

Theorem 5. *No polynomial-time algorithm exists for approximating the partition function $\mathcal{Z}$ of the zero-field, antiferromagnetic Ising model in neighborhoods of its Fisher zeros.*

5. Conclusion

Perhaps the foundational misconception of the P versus NP problem was, perhaps, that intractability boundaries are demarcated by time constraints as opposed to limitation on information bandwidth. The inability to further compress the $K(G \mid \mathbf{r}_\epsilon) \in \Theta(n)$ quantity of minimally sufficient information means that, upon asserting the conservation of complexity, the problem of approximating the partition function of the zero-field, antiferromagnetic Ising model exceeds the limitations of inductive inference: It is inferentially intractable and, as a consequence, computationally intractable.

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